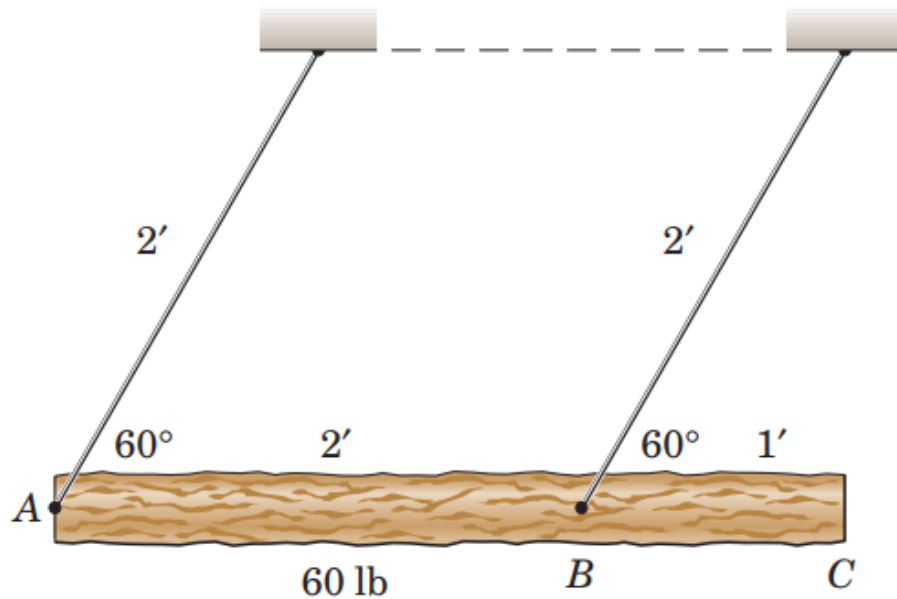


Kinetics of Rigid Bodies

Dr. A. Banerjee

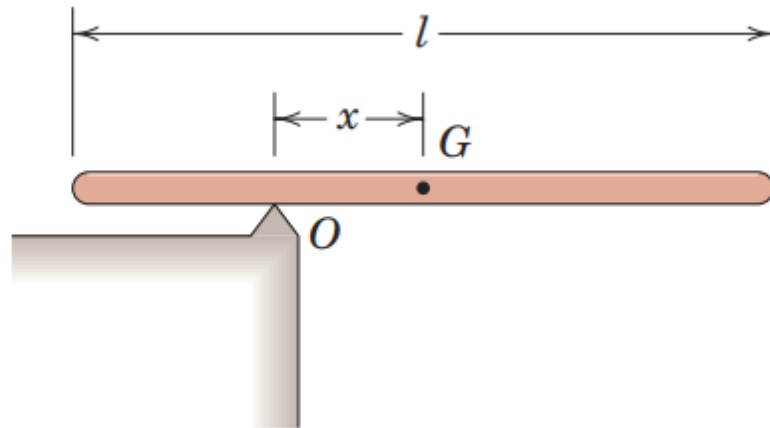
Example Problem

The uniform 60-lb log is supported by the two cables and used as a battering ram. If the log is released from rest in the position shown, calculate the initial tension induced in each cable immediately after release and the corresponding angular acceleration of the cables.



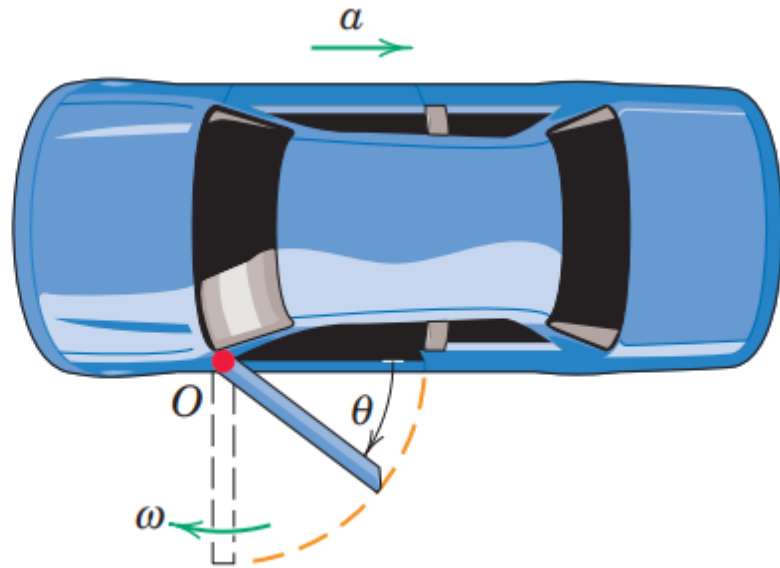
Example Problem

The uniform slender bar is released from rest in the horizontal position shown. Determine the value of x for which the angular acceleration is a maximum, and determine the corresponding angular acceleration α .



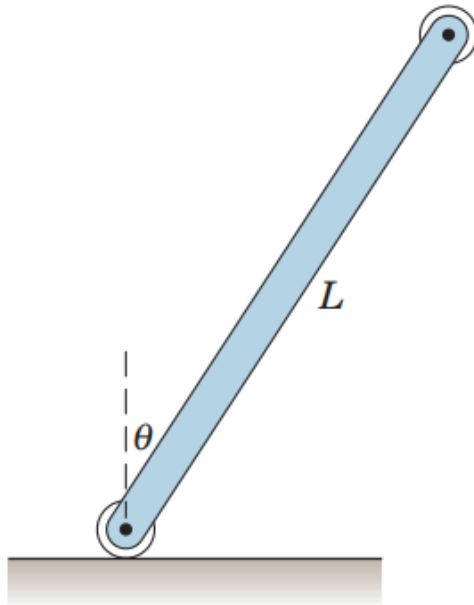
Example Problem

A car door is inadvertently left slightly open when the brakes are applied to give the car a constant rearward acceleration a . Derive expressions for the angular velocity of the door as it swings past the 90° position and the components of the hinge reactions for any value of θ . The mass of the door is m , its mass center is a distance $\bar{r}$ from the hinge axis O , and the radius of gyration about O is k_O .



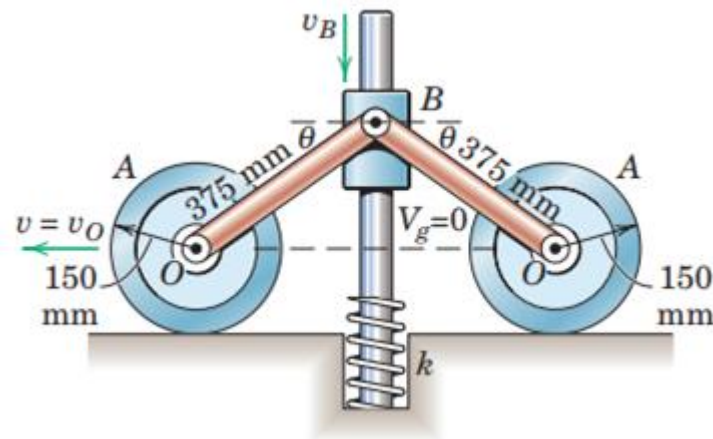
Example Problem

The uniform slender bar of mass m and length L with small end rollers is released from rest in the position shown with the lower roller in contact with the horizontal plane. Determine the normal force N under the lower roller and the angular acceleration α of the bar immediately after release.



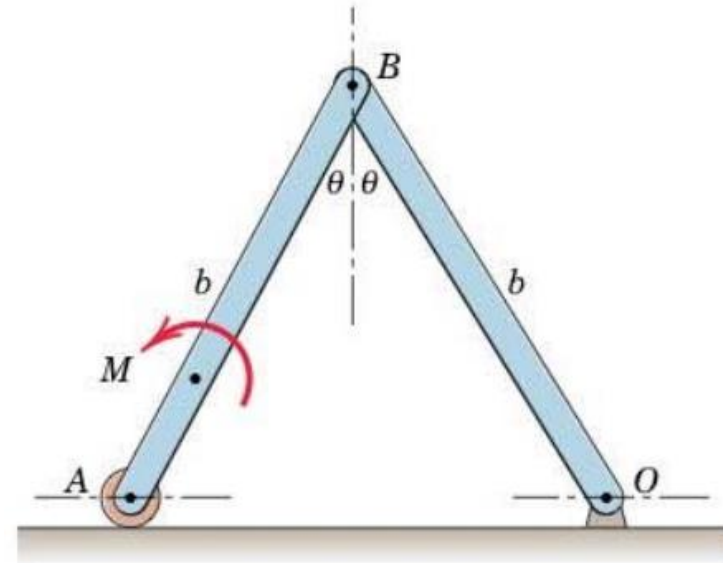
Example Problem

In the mechanism shown, each of the two wheels has a mass of 30 kg and a centroidal radius of gyration of 100 mm. Each link OB has a mass of 10 kg and may be treated as a slender bar. The 7-kg collar at B slides on the fixed vertical shaft with negligible friction. The spring has a stiffness $k = 30 \text{ kN/m}$ and is contacted by the bottom of the collar when the links reach the horizontal position. If the collar is released from rest at the position $\theta = 45^\circ$ and if friction is sufficient to prevent the wheels from slipping, determine (a) the velocity v_B of the collar as it first strikes the spring and (b) the maximum deformation x of the spring.



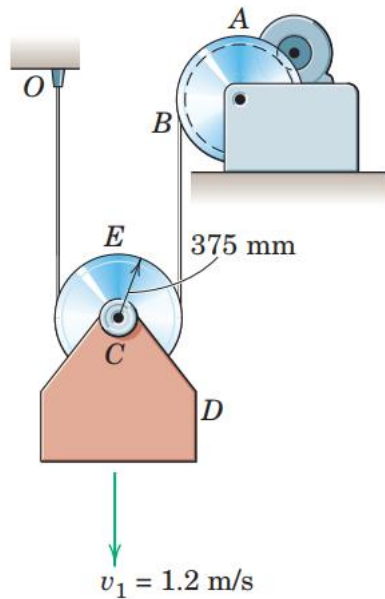
Example Problem

The two slender bars each of mass m and length b are pinned together and move in the vertical plane. If the bars are released from rest in the position shown and move together under the action of a couple M of constant magnitude applied to AB , determine the velocity of A as it strikes O .



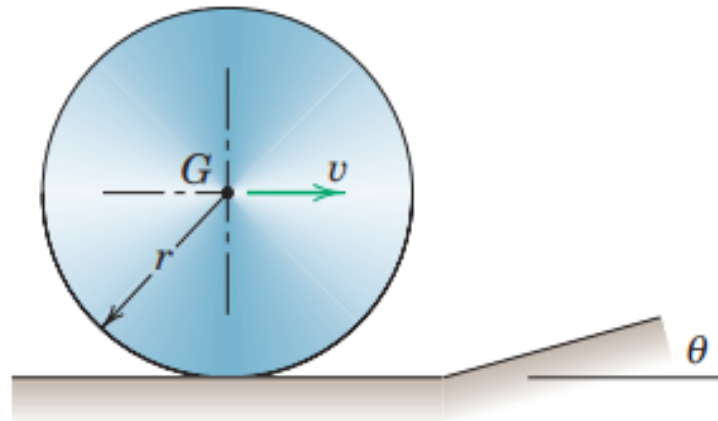
Example Problem

The sheave E of the hoisting rig shown has a mass of 30 kg and a centroidal radius of gyration of 250 mm. The 40-kg load D which is carried by the sheave has an initial downward velocity $v_1 = 1.2$ m/s at the instant when a clockwise torque is applied to the hoisting drum A to maintain essentially a constant force $F = 380$ N in the cable at B . Compute the angular velocity ω_2 of the sheave 5 seconds after the torque is applied to the drum and find the tension T in the cable at O during the interval. Neglect all friction.



Example Problem

A uniform circular disk which rolls with a velocity v without slipping encounters an abrupt change in the direction of its motion as it rolls onto the incline θ . Determine the new velocity v' of the center of the disk as it starts up the incline, and find the fraction n of the initial energy which is lost because of impact with the incline if $\theta = 10^\circ$.



Problem 6/198